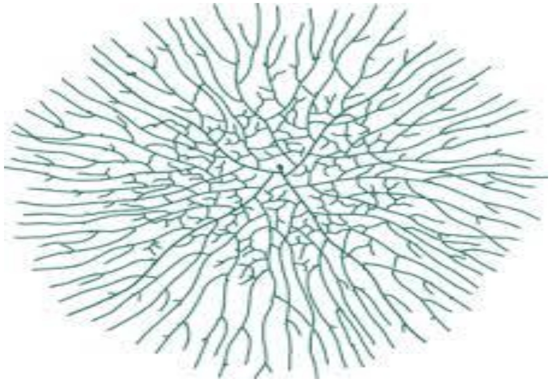
A fungus (plural: fungi) is any member of the group of eukaryotic organisms that includes **microorganisms such as yeasts and molds, as well as the more familiar mushrooms.** These organisms are classified as a **kingdom, fungi**, which is separate from the other eukaryotic life kingdoms of plants and animals.

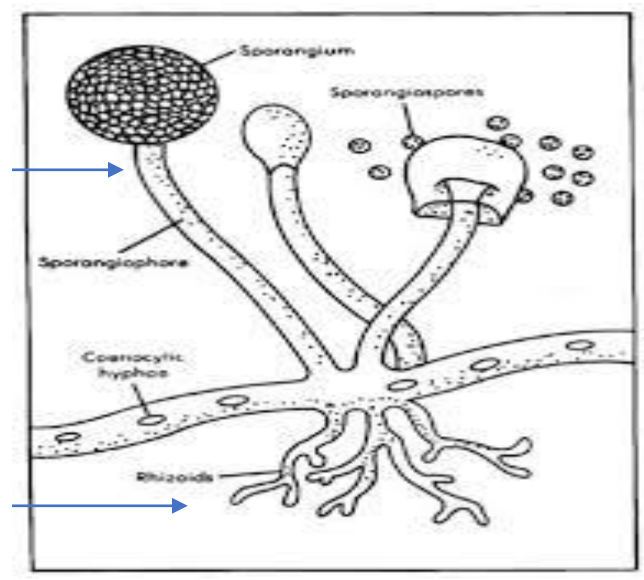
Fungi are widely varied that range from the smallest **unicellular fungi such as yeast** to **multicellular** capable of forming hyphal threads **as *Rhizopus*** that called **filamentous fungi**



A. filamentous fungi: Most filamentous fungi grow as hyphae. hyphae are cylindrical, thread-like structures 2–10 µm in diameter and up to several centimeters in length.

These hyphae are condenses to be a network called **mycelium**

There are two main types of mycelium depending on their functions:



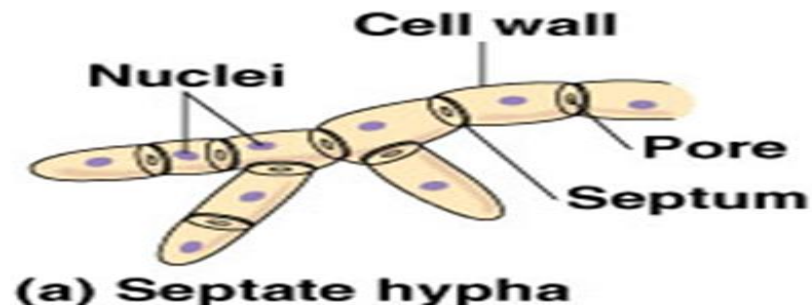
1. Aerial mycelium: which are the hyphae that are located above the food substance. When viewed closely under the microscope, aerial mycelium contain a spherical structure at the top of the hyphae. These are known as the **conidia** and sere as the **reproductive part** of the mold.

2. Vegetative (substrate) mycelium: which include the hyphae that penetrates into the substrate and absorbs nutrients for the continued growth and development of the organisms. These hyphae therefore act like plant roots.

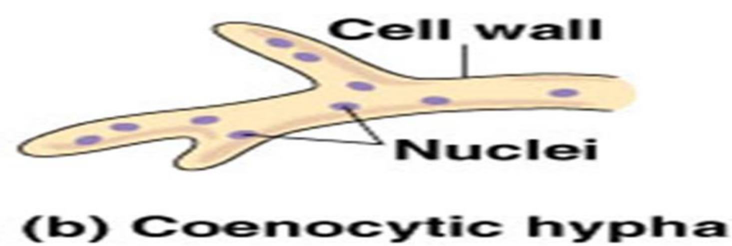
The vegetative hypha at the periphery of a colony is a tip-growing cellular element that undergoes regular branching, is commonly multinucleate, and usually produces septa.

Hyphae can be either septate or coenocytic.

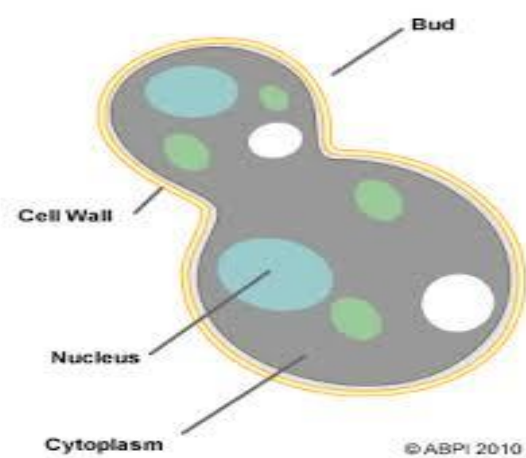
1. **Septate hyphae are divided into compartments separated by cross walls** (internal cell walls) called septa. Septa have pores that allow cytoplasm, organelles, and sometimes nuclei to pass through.



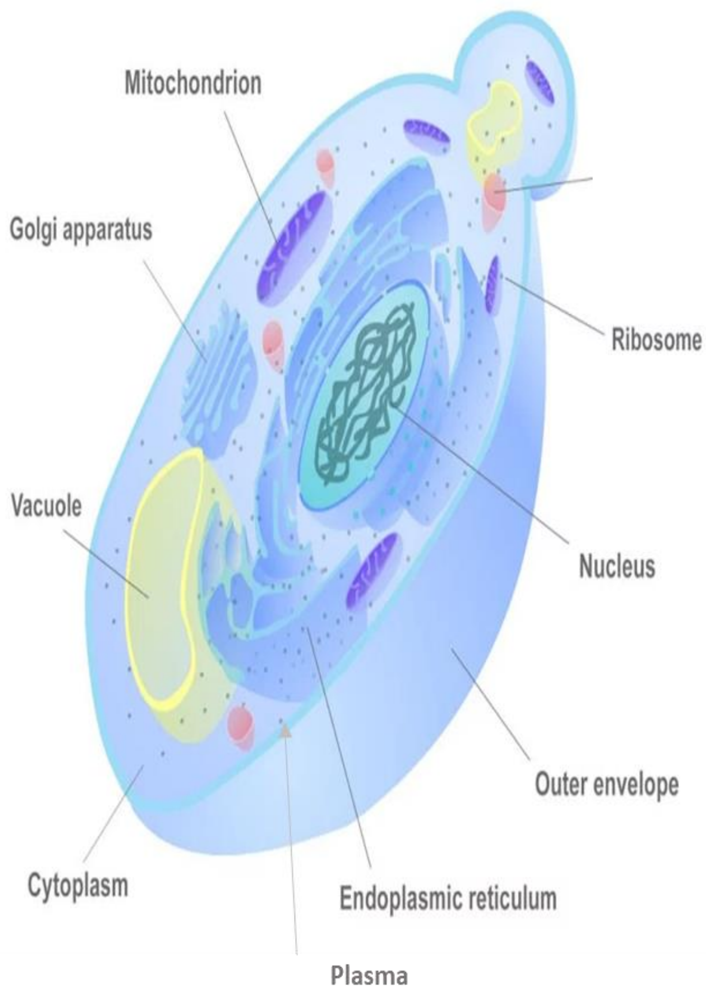
2. Coenocytic hyphae are not compartmentalized (i.e. not have septa). Coenocytic hyphae are **multinucleate** supercells.



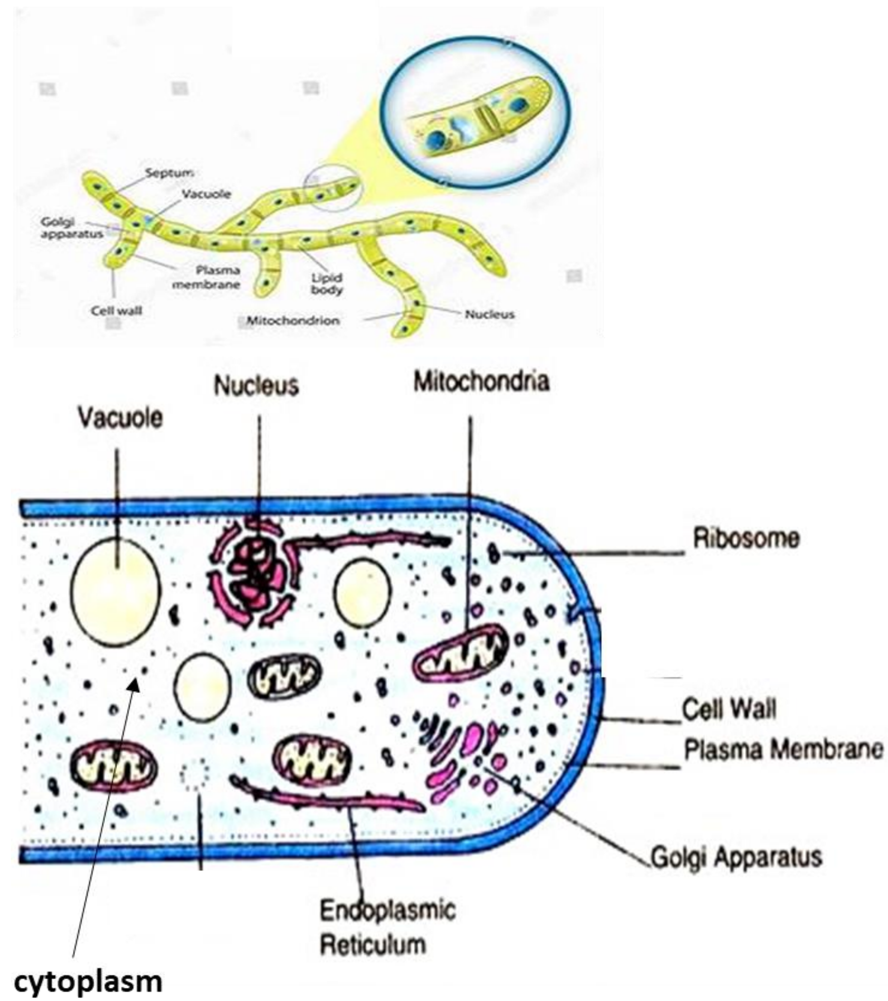
B. Unicellular fungi (Yeast) that divide asexually by budding or fission and whose individual cell size. It can vary widely from 2–3µm to 20–50µm in length and 1–10µm in width. ***S. cerevisiae***, the best known yeast, is generally ellipsoid in shape with a large diameter of 5–10µm and a small diameter of 1–7µm



1. Cell wall (cell envelope) **2. Cell membrane**
3.Cytoplasm **4. Nucleus**
5. Endoplasmic reticulum **6. Ribosome**
7. Mitochondria **8. Vacuole**
9. Golgi apparatus



Yeast cell structure



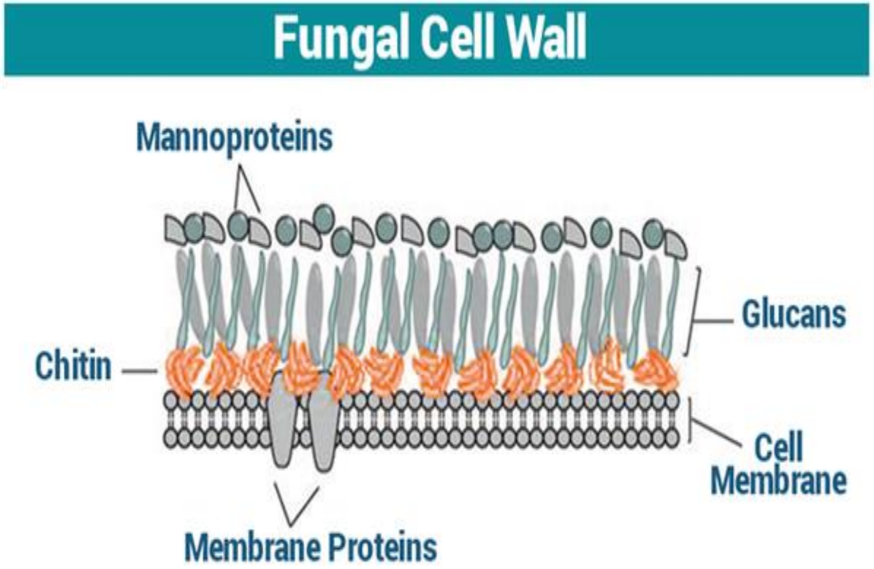
fungal cell structure

1. Cell wall (cell envelope)

Cell wall of fungi is a highly regulated, dynamic organelle surrounding the fungal cell and comprises the plasma membrane.

Fungal cell wall composed of a cross-linked network of 80-90% polysaccharides, and remaining proteins and lipids.

Polysaccharides as Chitin (a polymer of N-acetyl glucosamine), **glucans** are present in cell walls in the form of fibrils forming layers and in some cases **cellulose** (a polymer of D-glucose) also present.



The fungal cell wall serves numerous important essential roles including:

(1) determining and **maintaining the morphology** and integrity of developing fungal cells.

(2) **protecting fungal cells** against changes in external osmotic pressure and other environmental stresses;

(3) The cell wall **mediates the adhesion of cells** to one another and the substratum

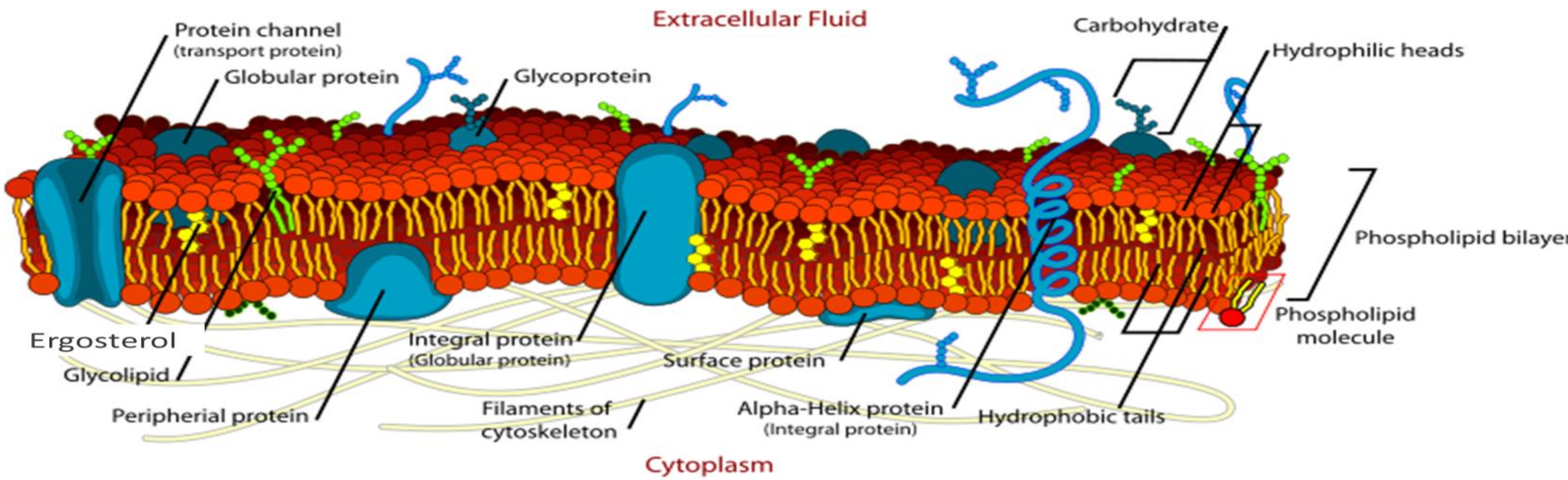
(4) serves as a **signaling center** to activate signal transduction pathways within the cell.

(5) **providing a dynamic interface** between the fungal cell and its surrounding environment.

2. Cell membrane (plasma membrane or Plasma lemma)

The plasma membrane of the fungal cell composed of **phospholipid bilayer** interspersed with **globular proteins** that regulate the entry of nutrients and exit of metabolites with structure similar to that of prokaryotes.

Ergosterol (lipid) is the major sterol found in the membranes of fungi, in contrast to the **cholesterol** found in the membranes of animals and **phytosterols** in plants.



Functions of the Plasma Membrane

A. Physical Barrier

The plasma membrane surrounds all cells and physically separates the cytoplasm, which is the material that makes up the cell, from the extracellular fluid outside the cell.

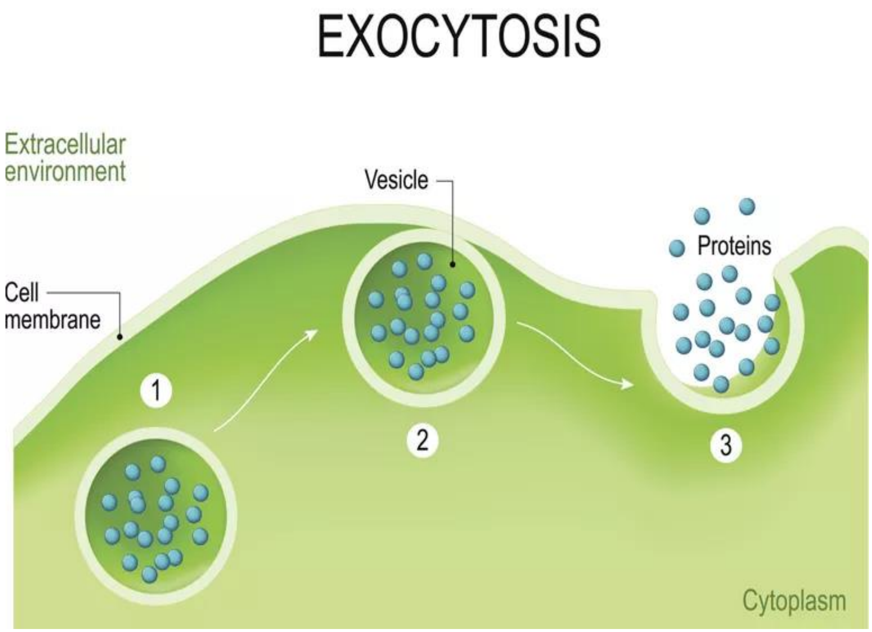
The plasma membrane provides structural support to the cell. It **tethers** the cytoskeleton, which is **a network of protein filaments** inside the cell that **hold all the parts of the cell in place**. This gives the cell its shape.

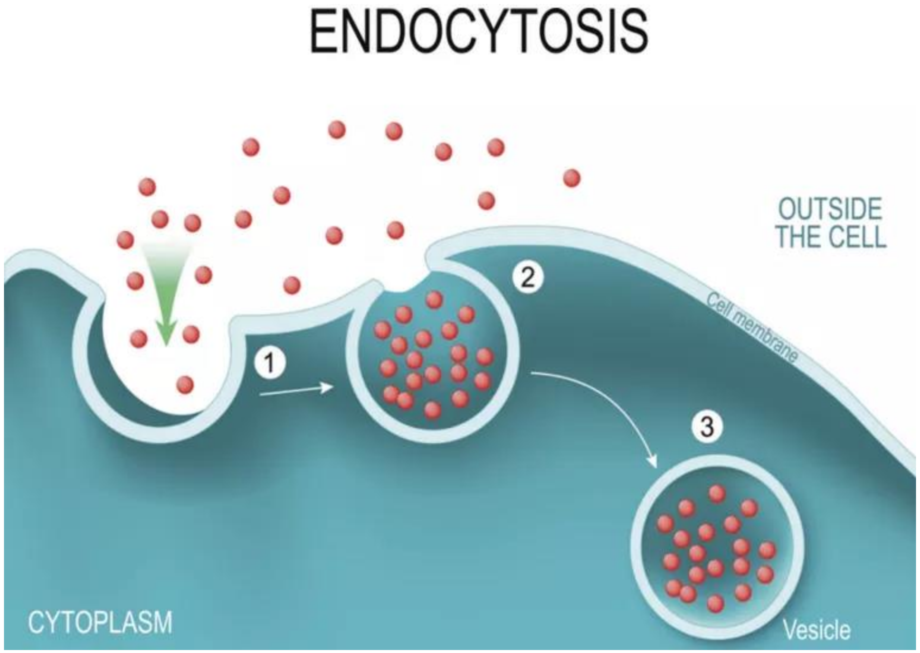
B. Selective Permeability

Plasma membranes are selectively permeable, meaning that only certain molecules can pass through them.

C. Endocytosis and Exocytosis

Endocytosis occur when a cell **ingests** contents relatively large (even whole bacteria from the extracellular fluid) than the molecules that pass through transport channels. **Exocytosis** occur when the cell releases these large materials. The cell membrane plays an important role in both of these processes. The shape of the membrane itself changes to allow molecules to enter or exit the cell. **It also forms vacuoles**, small bubbles of membrane that can transport many molecules to the cell.





3. Cytoplasm

Cytoplasm consists of all of the contents outside of the nucleus and enclosed within the cell membrane of a cell. It is **clear** in color and has a **gel-like appearance**. Cytoplasm is composed mainly of water but also contains enzymes, salts, organelles, and various organic molecules.

The cytoplasm consists of three main components. They are the **cytosol, organelles**, and various particles and granules called cytoplasmic **inclusions**.

1. Cytosol: is the semi-fluid or liquid medium of a cell's cytoplasm. It is located outside of the nucleus and within the cell membrane.

2. Organelles: Organelles are tiny cellular structures that perform specific functions within a cell. Examples of organelles include: mitochondria, ribosomes, nucleus, lysosomes, chloroplasts, endoplasmic reticulum, and Golgi apparatus. Also located within the cytoplasm is the cytoskeleton, a network of fibers that help the cell maintain its shape and provide support for organelles.

3. Cytoplasmic Inclusions: Cytoplasmic inclusions are particles that are temporarily suspended in the cytoplasm such as proteins, enzymes, Glycogen, Melanin, lipids and acids.

The cytoplasm can be divided into two primary parts: the endoplasm and ectoplasm. The **endoplasm is the central area** of the cytoplasm that contains the organelles. The **ectoplasm is the more gel-like peripheral portion** of the cytoplasm of a cell.

The cytoplasm functions to suspend organelles and cellular molecules. Many cellular **processes also occur in the cytoplasm**. Some of these processes include protein synthesis, the first stage of cellular respiration (known as glycolysis). In addition, the cytoplasm **helps to move materials, such as hormones**, around the cell and also dissolves cellular waste.

4. Nucleus

The cytoplasm of filamentous fungi contains **one, two or more globose or spherical nuclei** while **unicellular fungi have only one nucleus**. The **nucleus is the organelle which houses chromosomes.** Chromosomes consist of DNA, which contains heredity information for cell growth, development, and reproduction.

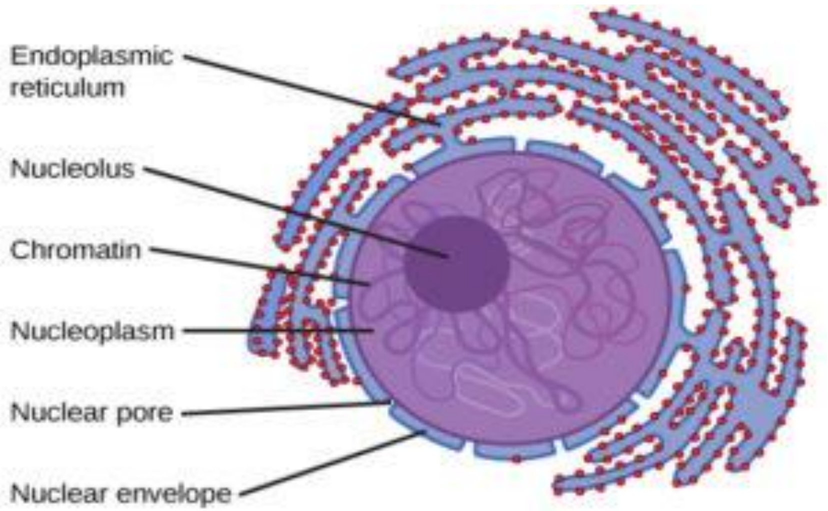
The cell nucleus is bound by a double membrane called the nuclear envelope. This membrane **separates the contents of the nucleus from the cytoplasm**. Like the cell membrane, the nuclear envelope consists of phospholipids bilayer. The envelope helps to **maintain the shape of the nucleus** and **assists in regulating the flow of molecules** into and out of the nucleus through nuclear pores.

Nucleoplasm is the gelatinous substance within the nuclear envelope. Also called karyoplasm, this semi-aqueous material is similar to cytoplasm and is composed mainly of water with dissolved salts, enzymes, and organic molecules suspended within. The nucleolus and chromosomes are surrounded by nucleoplasm, which protect the contents of the nucleus. Nucleoplasm also **supports the nucleus to maintain its shape.** Additionally, nucleoplasm **provides a medium by which materials**, such as enzymes and nucleotides (DNA and RNA subunits), **can be transported** throughout the nucleus.

The Nucleolus is a dense, membrane-less structure composed of RNA and proteins. The nucleolus contains **nucleolar organizers, which helps to synthesize ribosomes** by transcribing and assembling ribosomal RNA subunits. These subunits join together to form a ribosome during protein synthesis.

5. Endoplasmic reticulum

Presence of endoplasmic reticulum in fungal cytoplasm is observed through electron microscope. **It is made up of a system of microtubules beset with small granules.** In most of the fungi it is highly vesicular. It is loose and irregular as compared with cells of green plants. **In multinucleate hyphae the nuclei may be connected by endoplasmic reticulum.**

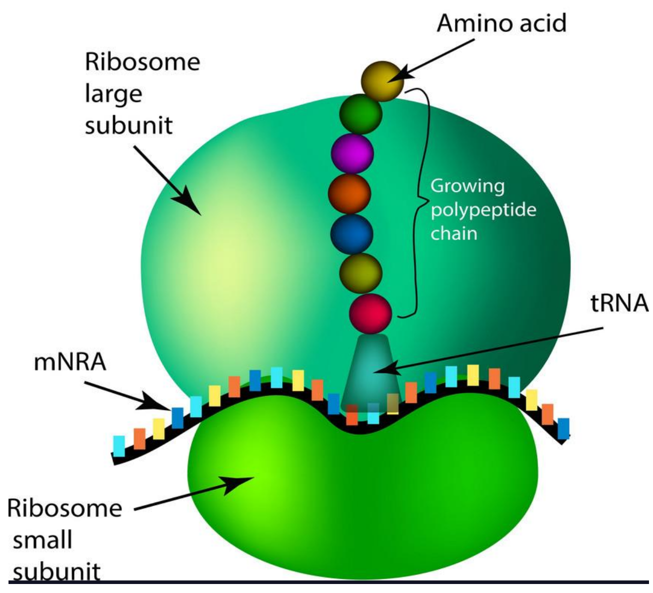


It plays a major role in the production, processing, and transport of proteins and lipids. The ER produces transmembrane proteins and lipids for its membrane and for many other cell components including the Golgi apparatus and the cell membrane.

6. Ribosome (80 S)

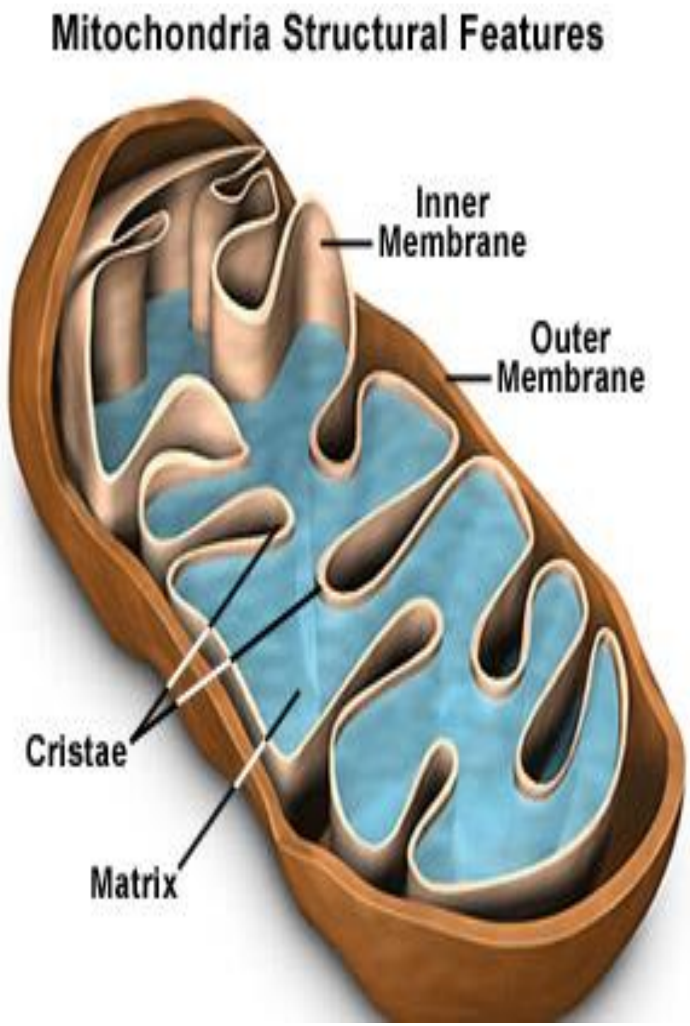
In eukaryotic cell ribosome may be **suspended in the cytosol or bound to the endoplasmic reticulum**. These ribosomes are called **free ribosomes and bound ribosomes** respectively.

Eukaryotic Ribosomal subunits (small 40 S and large 60 S) are synthesized in the nucleolus and cross over the nuclear membrane to the cytoplasm through nuclear pores. **Ribosome is responsible for protein synthesis**



7. Mitochondria

Numerous small and spherical to elongated bodies known as mitochondria are dispersed in cytoplasm. The number of mitochondria within a cell varies depending on the type and function of the cell.



Mitochondria are **covered by an outer double membrane;** The **inner membrane is folded** creating structures known as **cristae**. The folds enhance the "productivity" of cellular respiration by increasing the available surface area. Within the inner mitochondrial membrane are series of protein complexes and electron carrier molecules, which form **the electron transport chain (ETC).** The ETC represents the third stage of aerobic cellular respiration and the stage where the vast majority of ATP molecules are generated. So, mitochondria is known as power houses of eukaryotic cells.

The mitochondrial matrix contains mitochondrial DNA (mtDNA), ribosomes, and enzymes. Several of the steps in cellular respiration, including the Citric Acid Cycle and oxidative phosphorylation occur in the matrix due to its high concentration of enzymes There is no difference between mitochondria of fungi and green plants.

8. Vacuoles

Vacuoles are fluid-filled small vesicular, to larger spherical/ovoid structures found in the old cells of hyphae. The end of hyphal tip of **young hyphae lacks vacuole.** With the age, the vacuoles conjugate.

Functions of vacuoles:

1. Turgor pressure control: The water filled central vacuole exerts pressure on the cell wall to maintain the cell shape.

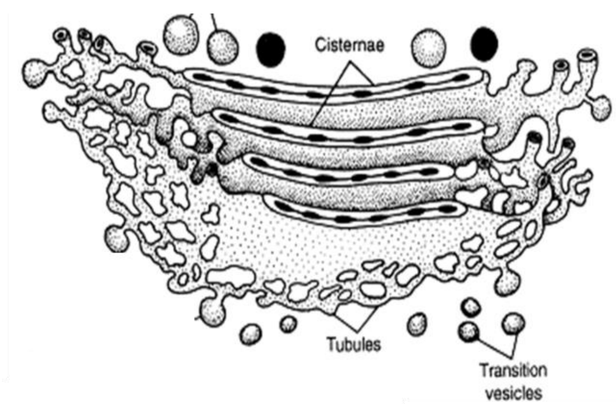
2. Storage: vacuoles store important minerals, water, nutrients, ions, waste products, small molecules, enzymes, and plant pigments.

3. Molecule degradation: the internal acidic environment of a vacuole aids in the degradation of larger molecules sent to the vacuole for destruction. In one type of autophagy (macro autophagy), cytoplasm and organelles are sequestered in double-membrane bounded vesicles called **autophagosomes**, which with vacuoles, allowing their contents to be broken down by vacuolar hydrolases.

9. Golgi apparatus

Golgi apparatus is composed of membrane-bound sacs called cisternae which are **rare** found in fungal cells except in Oomycetes.

The job of the Golgi apparatus is to process and bundle macromolecules like proteins and lipids as they are synthesized within the cell. The Golgi apparatus is sometimes **compared to a post office inside the cell** since one major function is to modify, sort, and package proteins to be secreted.



Nutrition of fungi

Fungi need some elements which are essential for their growth. These elements may be **Macro Elements as C, N, O, H, S, P** which are are body builders and provide energy for metabolic processes or **microelements as K, Mg, S, P, Mn, Cu, Mo, Fe and Zn.** The chief sources of **carbon** for most fungi are the **carbohydrates (organic)**. The carbohydrates are needed for building up the body and also as a source of energy.

Besides carbon, **fungi require nitrogen**. To obtain nitrogen, **they utilize both organic and inorganic materials**. The chief organic sources of **nitrogen are protein, peptide or an amino acid.** Many fungi, obtain nitrogen from **inorganic sources as nitrate and ammonium salts.**

Hydrogen and oxygen are supplied in the form of water which is the major constituent of fungus mycelium forming about 85-90% of the entire weight.

Also, **fungi require sulphur, phosphorus in fairly large amounts for their nutrition** which are obtained from simple **inorganic salts** such as suIphates for sulphur, and phosphates for phosphorus.

Some fungi are reported to require only minute traces of iron, zinc, copper, manganese and cobalt and molybdenum. These trace elements or micronutrients are considered important of growth.

The fungi like all other organisms require minute amounts of specific, relatively complex organic compounds for growth such as vitamins. Many fungi synthesize their own supply of growth factors from a simple nutrient solution of defined composition. There are others which depend in whole or in part on an external source because they are unable to synthesize one or more of the essential growth substances.

Besides the nutritional requirements listed above the growth of fungi is **habitat factors such as temperature, oxygen supply, moisture, pH value and by-products of metabolism.**

Mode of nutrition

Fungi are heterotrophic in nutrition. They are chlorophyll deficient organisms and hence they **cannot manufacture carbohydrates using carbon dioxide, water and sunlight as plant**. therefore they are always dependent for their food on some dead organic material or living beings.

According to their method of obtaining food are divided into three categories that are:

1. Saprophytes 2. Parasites 3. Symbionts

1. Saprophytes : The saprophytic fungi grow upon dead organic matters such as rotten fruits, rotten vegetables, moist wood, moist leather, pickles, cheese, rotting leaves, plant debris, manures, moist bread and many other possible dead organic materials.

The saprophytic fungi absorb their food from the substratum by ordinary vegetative hyphae which penetrate the substratum, e.g., *Mucor mucedo*. In other cases of the saprophytic fungi such as *Rhizopus* the rhizoids develop which penetrate the substratum and absorb the food material.

- **Saprophytic fungi produce exoenzymes** (enzymes which acts outside the cell)
- These enzymes digest the complex organic matter in the substratum into simpler compounds to facilitate easy absorption by the hyphae.

- **Saprophytic fungi may be of two types:-**

1. **Ectophytic saprophytes**: grown on the surface of organic matter
2. **Endophytic saprophytes**: grown inside the organic matter

2. Parasites: The parasitic fungi absorb their food material from the living tissues of the hosts on which they parasitize. Such parasitic fungi are quite harmful to their hosts and **cause many serious diseases**. Some of these fungi can cause the great losses to the **human** beings. Many diseases of the important **crops** are caused by parasitic fungi. The rusts, smuts and many other plant diseases are important examples of fungal diseases of crops. Their mode of life is parasitic and the relation of host and parasite is called the **parasitism.**

According to the degree of parasitism the parasitic fungi divided into:

1.Obligate parasites: these fungi can live only as parasite on a living host. Obligate parasites cannot live on dead organic matter Example: ***Puccinia*** which cause rust disease in several crop plants including wheat

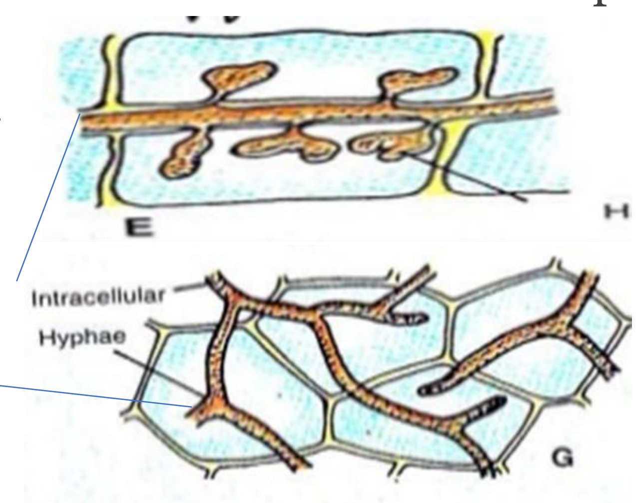
2.Facultative saprophytes: They are parasites, but they can also survive on dead organic matter in the absence of living host Example: **Taphrina.**

3.Facultative parasites: these fungi usually follow saprophytic mode of nutrition but under certain conditions, they parasitize suitable host plants Example: **Fusarium and Pythium** which cause soft rot disease in crop plants.

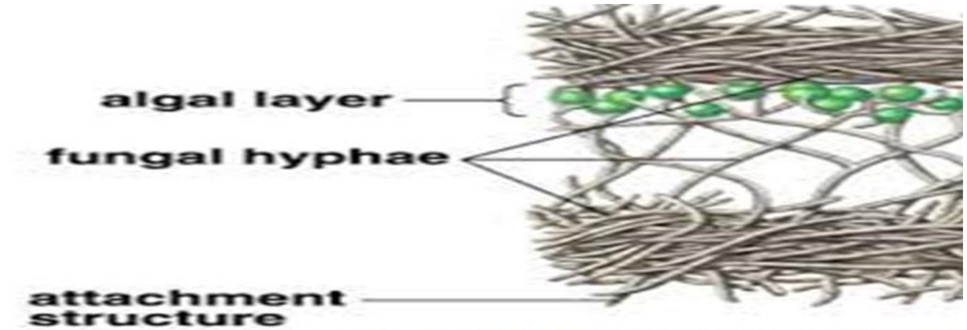
•Parasitic fungi possess specialized **absorptive structures called haustoria** for the absorption of nutrients from the host cells.

•Haustoria are specialized hyphal modifications which may be **inter-cellular** (occupy between two cells) in **intra-cellular** (occupy within the cell)

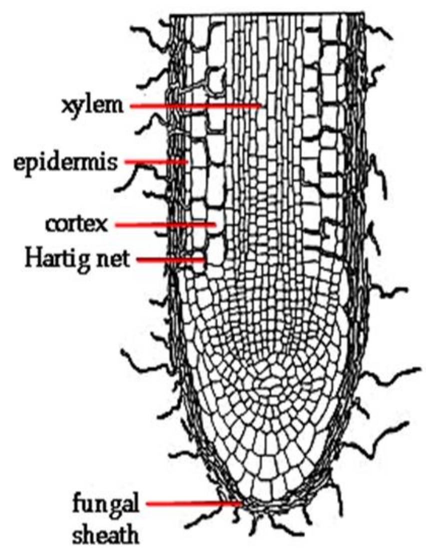
•Size and shape of haustoria varies in different fungal groups.

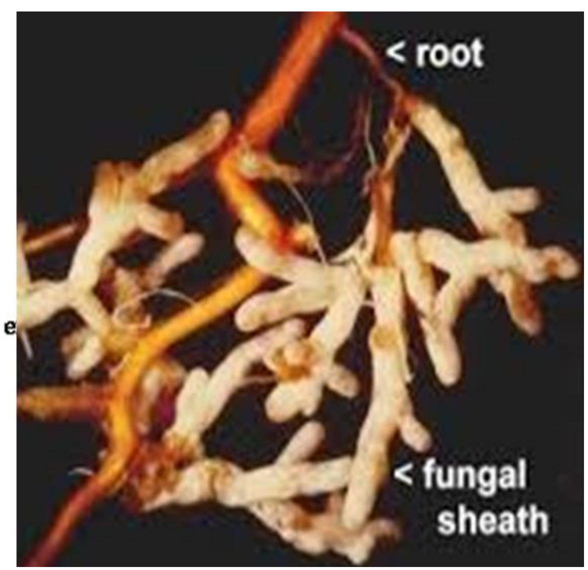


3. symbionts: Some fungi live in **closed relationship with other organism** where they are mutually beneficial to each other. Such relationship is called the '**symbiosis**' and the participants the '**symbionts**'. The most striking examples are the **lichens and mycorrhiza**.



The lichens are the resultants of the symbiotic association of **algae and fungi.** Here, the fungal partner is responsible for the fixation and absorption of inorganic nutrients and water and the algal partner synthesizes the organic food.





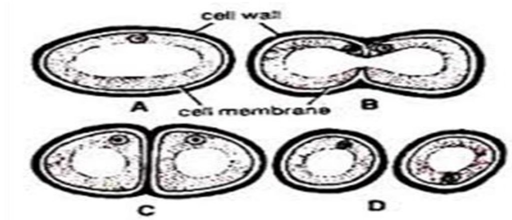
Certain **fungi** develop in the roots of **higher plants** and the **mycorrhiza** are developed. Here the fungi absorb their food from the roots where the fungi develop a hyphal network which increase the surface area through which it can decompose the organic matters like plant debris by extracellular enzymes (cellulose, starch and other polymer) to more simple compounds which more available to the plant root.

Reproduction in Fungi

Some of the important methods of reproduction in Fungi are as follows:

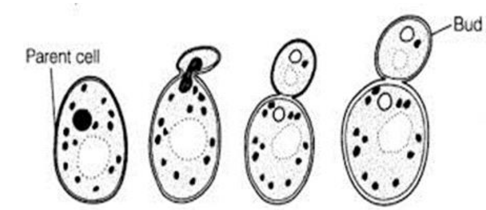
1. Vegetative reproduction:

The most common method of vegetative reproduction is fragmentation. The hypha breaks up into small fragments accidentally or otherwise. Each fragment develops into a new individual. In the laboratory the 'hyphal tip method' is commonly used for inoculation of saprophytic fungus.



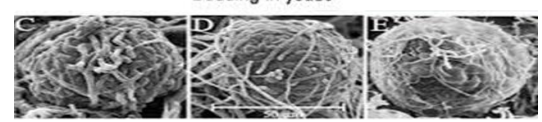
Fission in fungi

In addition to above-mentioned common method of vegetative reproduction the fungi reproduced vegetatively by other means, such as **fission, budding, sclerotia, rhizomorphs, etc**. **In fission**, the cell constricts in the center and divides into two giving rise to new individuals.



Budding in yeast

The budding is commonly found in Saccharomyces. The buds arise from the protoplasm of the parent cells and ultimately become new individuals.



The sclerotia are resistant and perennating bodies. They survive for many years. Each sclerotium is cushion-like structure of compact mycelium. They give rise to new mycelia on the approach of favorable conditions.

The rhizomorphs are long strands of parallel hyphae cemented together. it also resistant to unfavorable conditions and give rise to new mycelia even after several years on the approach of favorable conditions.

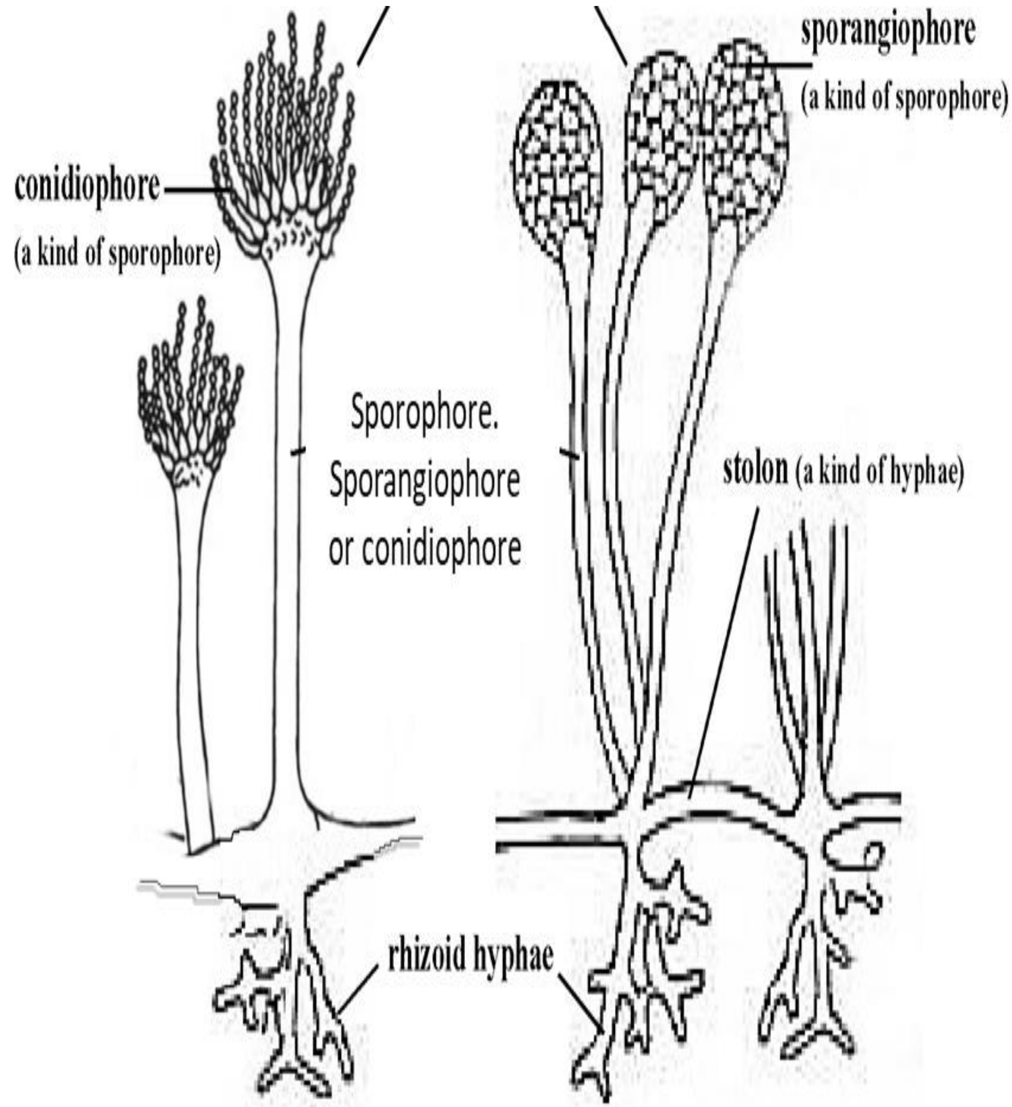
2. Asexual reproduction:

The asexual reproduction takes place by means of spores. Each spore may develop into a new individual.

- **Asexual spores:**

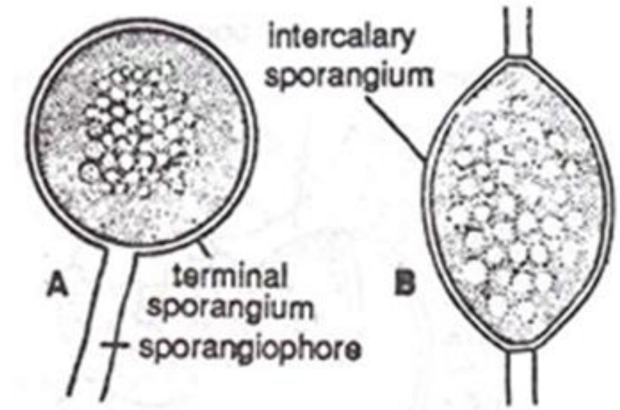
The spores are of **diverse type** and borne upon special structures called the **sporophores.**

The fungus producing more than one type of spores is called the pleomorphic or polymorphic. The spores produced inside the sporangia are termed the **endogenous spores** and the spores developing exogenously on the terminal ends of sporophores are called the **exogenous spores**.

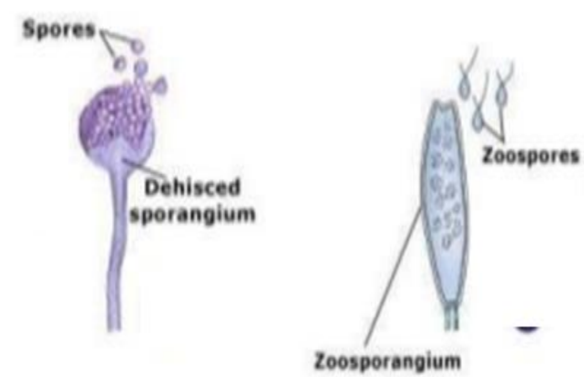


- **Endogenous spores:**

The endogenous spores are produced within the special spore producing cell called sporangium. The sporangia may be **terminal or intercalary** in their position. **The sporophores which bear the sporangia on their apices are called the sporangiophores.**



The spores produced inside the sporangia are called the endospores or endogenous spores. They may be **motile or non-motile**. The motile spores are called the **zoospores** and the non-motile **aplanospores**. The protoplasm of the sporangium divides into uninucleate or multinucleate protoplasmic bits and each bit transforms into a spore.

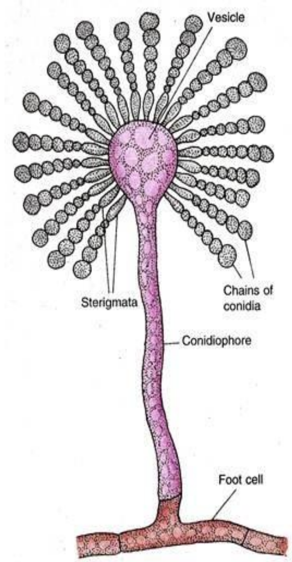


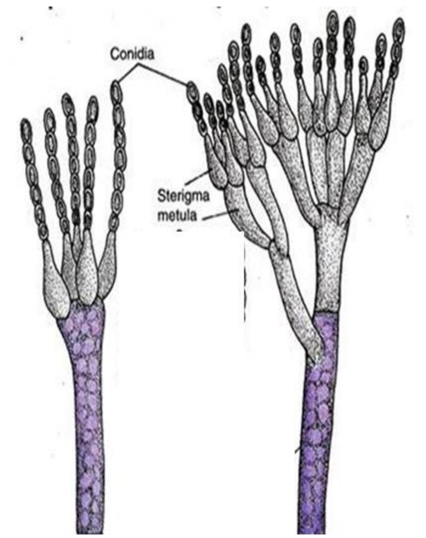
The endogenously produced **zoospores are usually kidney-shaped move with the help of their flagella (uni or biflagellate) the flagella are inserted posteriorly or laterally on them, zoospores without any cell wall, uninucleate and vacuolate**. Such zoospores have been recorded from *Albugo*, *Pythium*, *Phytophthora* and many other lower fungi.

The aplanospores are non-motile, without flagella and formed inside the sporangia. They may be uni or multinucleate (e. g., *Mucor, Rhizopus*). These spores **lack vacuoles and possess two layered cell walls**.

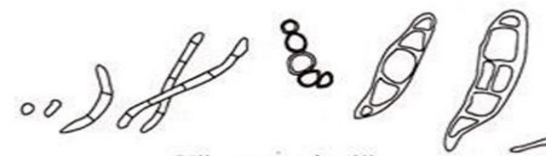
- **Exogenous spores:**

The spores producing externally are either called the exogenous spores or conidia. They are produced externally on the branched or unbranched **conidiophores.** The condiophores may **be septate or aseptate**. The conidia borne upon the **terminal apices** of the conidiophores or the ends of the branches of the condiophores which called strigmata.





The conidia may be produced singly or in chains. They may be unicellular or multicellular, uninucleate or multinucleate and diverse in their shape, color and size.

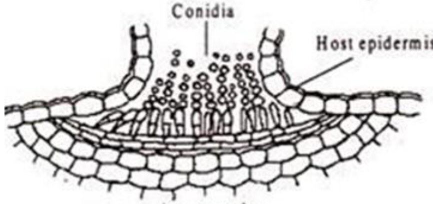


Different types of conidia

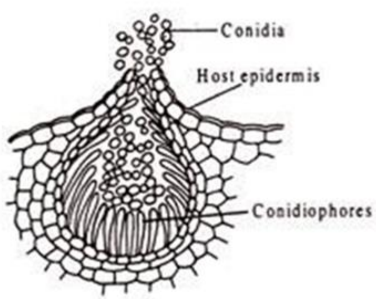
In other type of exospores, the sporophores develop in groups and form the specialized structure such as the pycnia and acervuli.

The pycnia are flask-shaped producing pycniospores in them.

The acervuli are saucer- shaped widely open bodies having developed conidia in them on small conidiophores.

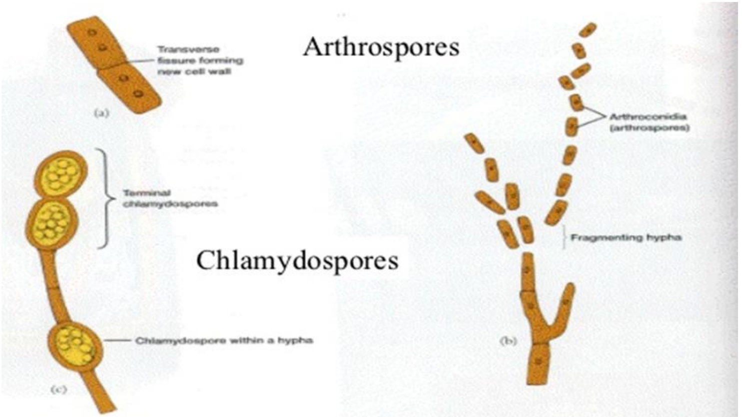


An acervulus.
Conidiophores organised in a bed on the host plant



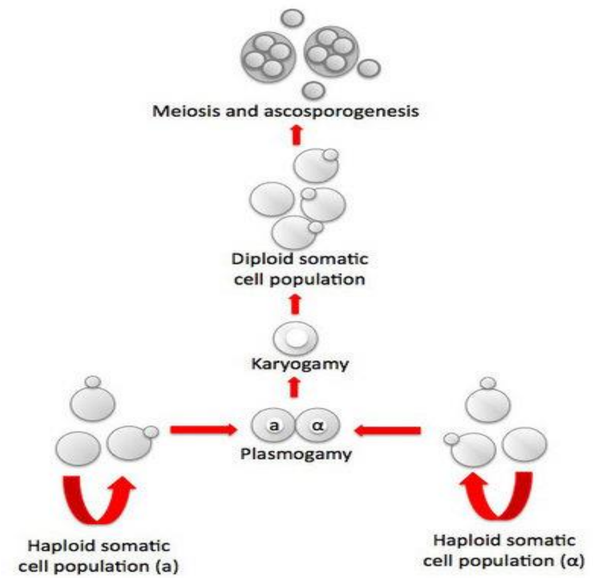
A pycnidium.
An organised flask-shaped structure embedded in host tissue.
The conidiophores are produced inside the pycnidium.

- **There are different types of asexual spores such as:**
- **Chlamydospores:** terminal parts of hyphae enveloped in wall and develop into resting spores.
- **Arthrospores:** hyphae break up into a component cells which behave as spores.

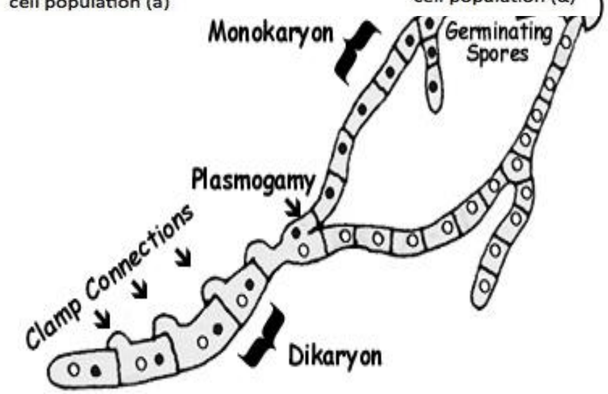


3. Sexual reproduction:

Sexual reproduction takes place by fusion of two nuclei originating from two individuals of opposite mating types, generally designated as male and female. Fusion started with protoplasts; the process is called **plasmogamy followed by** Fusion of the nuclei which known **as karyogamy.** It leads to production of a diploid cell, called **zygote. Finally, meiosis** takes place to restore the **hyploid cells.**



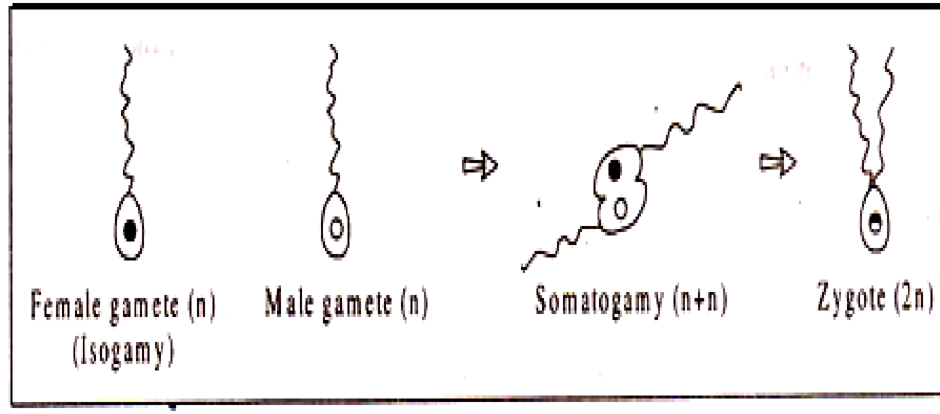
In the higher fungi, i.e. in ascomycetes and basidiomycetes, karyogamy does not follow **plasmogamy immediately. the fusion of the two nuclei of different strains is delayed** and the pairs of the nuclei **called the 'dicaryons**' are formed. The mycelium having such pairs of nuclei is called the **'dicaryotic mycelium'.**



The most common methods of sexual reproduction are as follows:

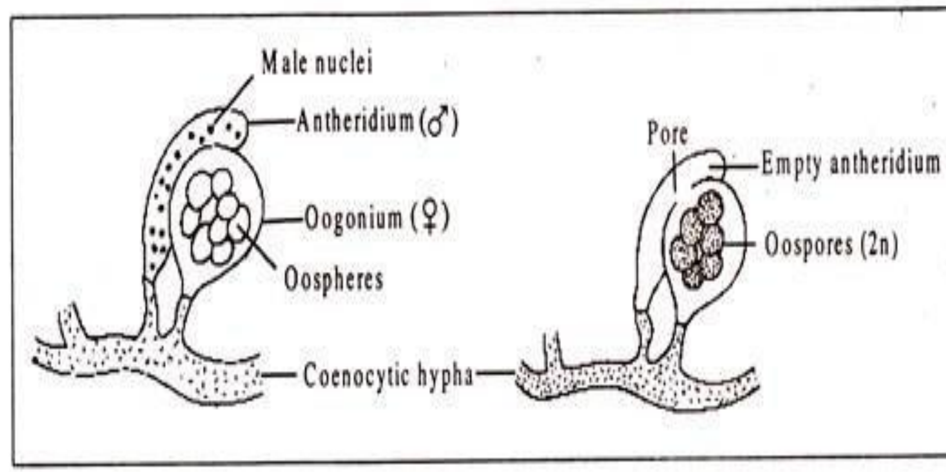
i. Planogametic copulation:

This type of sexual reproduction involves the **fusion of two naked gametes one or both of them are motile.** The **motile gametes are known as planogametes**. As in *Synchytrium, Plasmodiophora* etc.



ii. Gametangial contact:

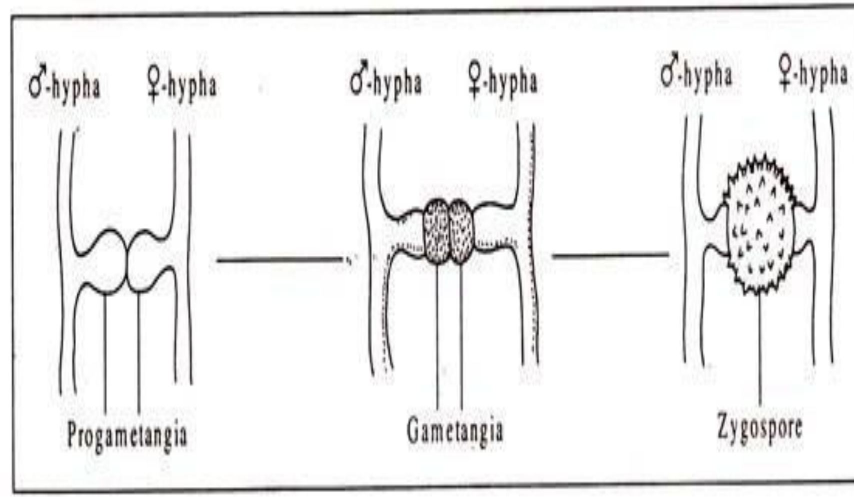
In this method two gametangia of opposite sex (oogonium and antheridium) come in contact and the nuclei of the male gametangium (antheridium) migrate to the female gametangium (oogonium). In no case the gametangia actually fuse.



The male nuclei in some species enter the female gametangium **through a pore** developed by the **dissolution of the wall of contact** (e.g., in *Aspergillus, Penicillium*, etc.); in other species the male nuclei migrate **through a fertilization tube** (e.g., *Phythium, Albugo*, etc.). **After the migration** of the nuclei the antheridium eventually break down but the oogonium continues its development in various ways to produce diploid oospores..

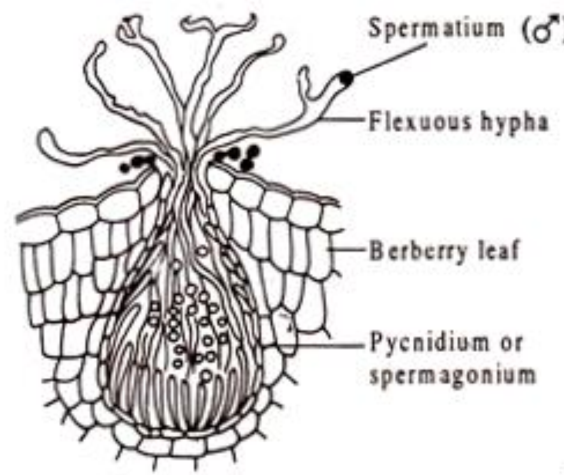
iii. Gametangial copulation:

In this method of sexual reproduction the fusion of the entire contents of two contacting compatible gametangia takes place **(e.g., *Mucor*, *Rhizopus*, etc**.) Thereby, their separate identity is lost and the fusion produces a zygospore. In the zygospore, the male and female nuclei pair with each other. Eventually, some of these pairs of nuclei fuse and the diploid nuclei undergo meiosis to restore haploidy.



iv. Spermatization:

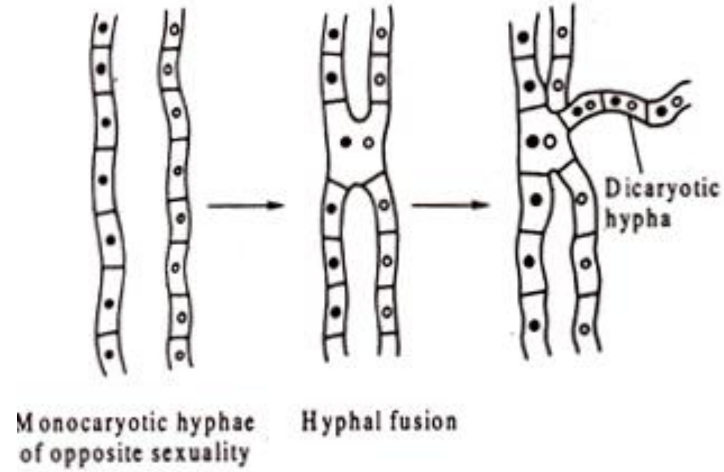
In some fungi such as the rust-fungus *Puccinia*, the male gametes are called spermatia which are produced in pycnidia (or spermogonia). **The minute, uninucleate, spore-like male structures** are passively transferred by insects to the receptive female hyphae. **A pore develops at the wall of contact** and the contents of spermatium pass into the female spermogonia through the receptive hypha leading to plasmogamy followed by Karyogamy which leads to formation of a diploid nucleus which undergoes meiosis to produce the different types of sexual spores.



Spermatization in *Puccinia graminis*. Spermatia from one spermagonium is transferred by insects to receptive (flexuous) hypha of another spermagonium.

v. Somatogamy:

In higher basidiomycetes, **like mushrooms, specialized sex organs are totally absent**. In these fungi, **fusion of vegetative hyphae originating from mycelia of different types occurred**.



Somatogamy in *Psalliota campestris*

The dikaryotic mycelium grows indefinitely and may, under appropriate conditions, lead to **formation of a fruit-body (basidiocarp).** In some of the ultimate cells of the dikaryotic hyphae constituting the basidiocarp, the pairs of nuclei fuse to produce diploid nuclei which undergo meiotic division and produce **haploid basidiospores**. The basidiospores on liberation germinate to produce **haploid mycelia** of different mating types